

PART 1: INTRODUCTION TO LOGS

Research and explain the following:

- 1. What are Logs?**
- 2. Why are logs important in Cyber Security?**
- 3. Types of Logs:**
 - **System Logs**
 - **Application Logs**
 - **Web Server Logs**
 - **Authentication Logs**
 - **Firewall Logs**

What are Logs?

Logs are automatically generated records that document events and activities occurring within computer systems, applications, servers, and network devices. They provide a chronological history of system operations and user actions. Logs contain valuable information such as login attempts, system errors, file access activities, network connections, security alerts, and application events. These records help administrators and security professionals understand what has happened on a system and identify any unusual or suspicious activities.

Why are Logs Important in Cyber Security?

Logs play a crucial role in cyber security because they provide visibility into system and network activities. They help organizations detect security incidents, investigate attacks, and monitor system health. By analyzing logs, security teams can identify unauthorized access attempts, malware infections, policy violations, and other threats.

Importance of Logs in Cyber Security

1. **Threat Detection:** Logs help identify suspicious activities such as repeated failed login attempts, unauthorized access, and malware behavior.
2. **Incident Investigation:** Security analysts use logs to reconstruct events and determine the cause and impact of security incidents.
3. **Evidence Collection:** Logs serve as digital evidence during forensic investigations and legal proceedings.
4. **Compliance Requirements:** Many security standards and regulations require organizations to maintain and review logs regularly.
5. **System Monitoring:** Logs help administrators monitor system performance, detect errors, and ensure normal operations.
6. **Audit Trails:** Logs provide a record of user actions and system events, helping organizations maintain accountability.

Types of Logs

1. System Logs

System logs record events related to the operating system and hardware components. These logs provide information about system startup and shutdown processes, device driver activities, service status changes, hardware failures, and system errors.

Examples:

- Operating system startup and shutdown events
- Hardware and driver errors
- Service and process activities
- System configuration changes

Purpose:

System logs help administrators monitor system performance, troubleshoot technical issues, and identify system failures.

2. Application Logs

Application logs are generated by software applications and record events related to application operations. These logs contain information about user activities, application errors, warnings, transactions, and software performance.

Examples:

- User actions within an application
- Application startup and shutdown events
- Software errors and exceptions
- Database transactions

Purpose:

Application logs assist developers and administrators in diagnosing software issues, monitoring application performance, and identifying security-related events.

3. Web Server Logs

Web server logs record all requests made to a web server. These logs provide details about website visitors, requested resources, response codes, and browser information.

Examples:

- Client IP addresses
- Requested web pages and files
- HTTP request methods (GET, POST)
- Response status codes (200, 404, 500)

- Browser and operating system details

Purpose:

Web server logs help administrators analyze website traffic, monitor user behavior, troubleshoot web application issues, and detect malicious activities such as scanning and attack attempts.

4. Authentication Logs

Authentication logs record user login and logout activities and other authentication-related events. These logs are essential for monitoring access to systems and applications.

Examples:

- Successful login attempts
- Failed login attempts
- User logout events
- Password changes
- Account lockouts

Purpose:

Authentication logs help detect unauthorized access attempts, brute-force attacks, credential misuse, and account compromise incidents.

5. Firewall Logs

Firewall logs record network traffic that is allowed or blocked by a firewall. They provide detailed information about incoming and outgoing connections and security policy enforcement.

Examples:

- Blocked connection attempts
- Allowed network traffic
- Source and destination IP addresses
- Port numbers and protocols used
- Security rule matches

Purpose:

Firewall logs help security teams identify malicious network activities, unauthorized access attempts, port scanning attacks, and other network-based threats.

PART 2: WINDOWS EVENT VIEWER ANALYSIS

Open Event Viewer on Windows.

Go to:

Windows Logs → Security

Analyze and record:

1. Total number of events.
2. Latest login event.
3. Latest logout event.
4. Failed login attempts (if any).
5. Event ID of login events.

Take screenshots.

Objective

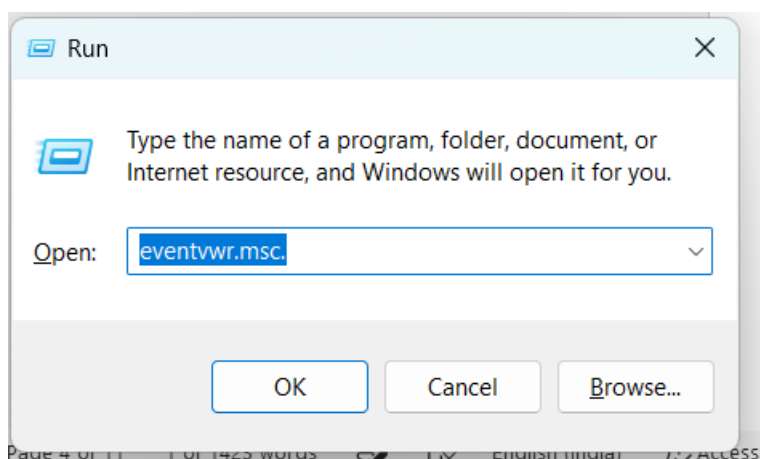
To analyze Windows Security Logs using Event Viewer and identify successful login events, failed login attempts, and logout events.

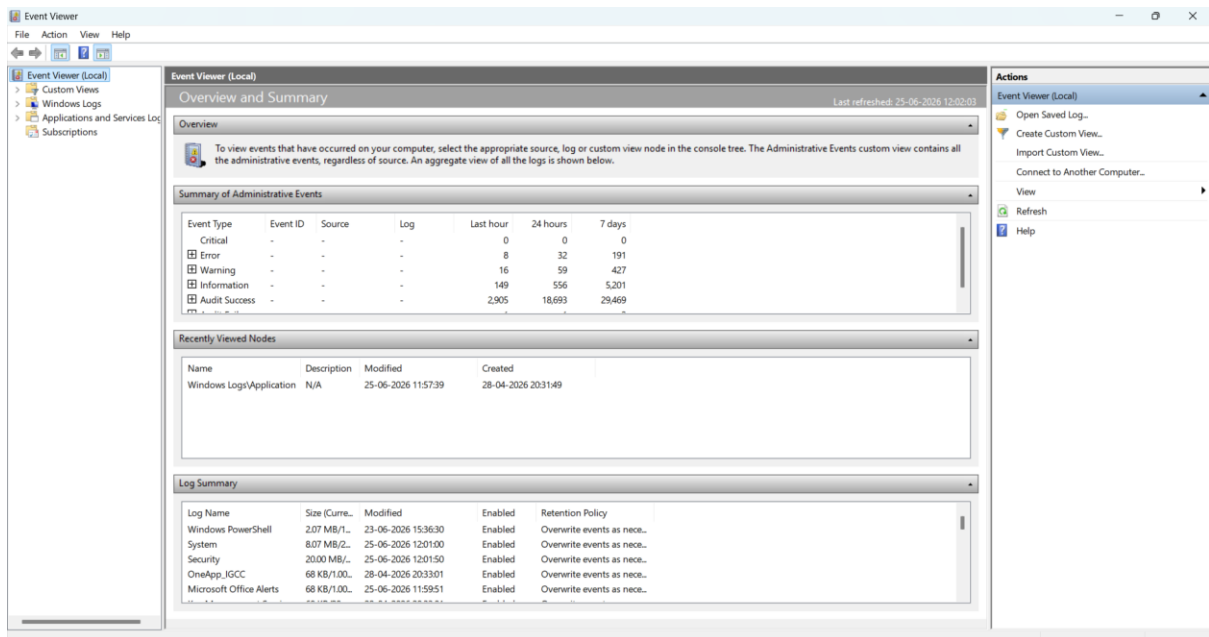
Step 1: Open Event Viewer

Procedure

1. Press **Windows + R**.
2. Type **eventvwr.msc**.
3. Click **OK** or press **Enter**.

Screenshot





Step 2: Access Security Logs

Procedure

1. In the left panel, expand **Windows Logs**.
2. Click **Security**.

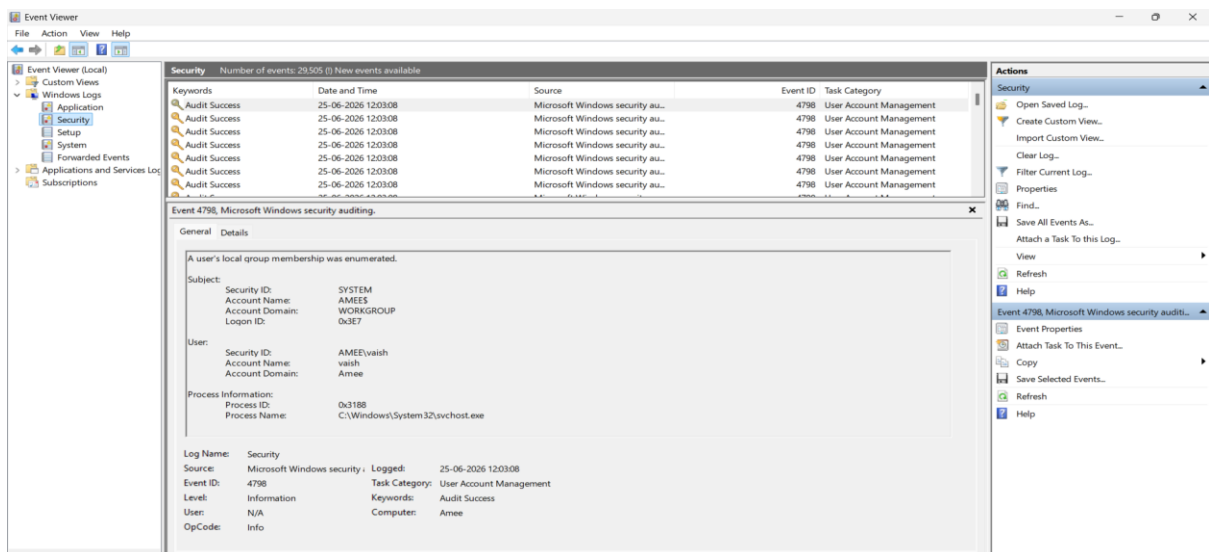
Path:

Event Viewer

└─ Windows Logs

└─ Security

Screenshot



Step 3: Identify Successful Login Event

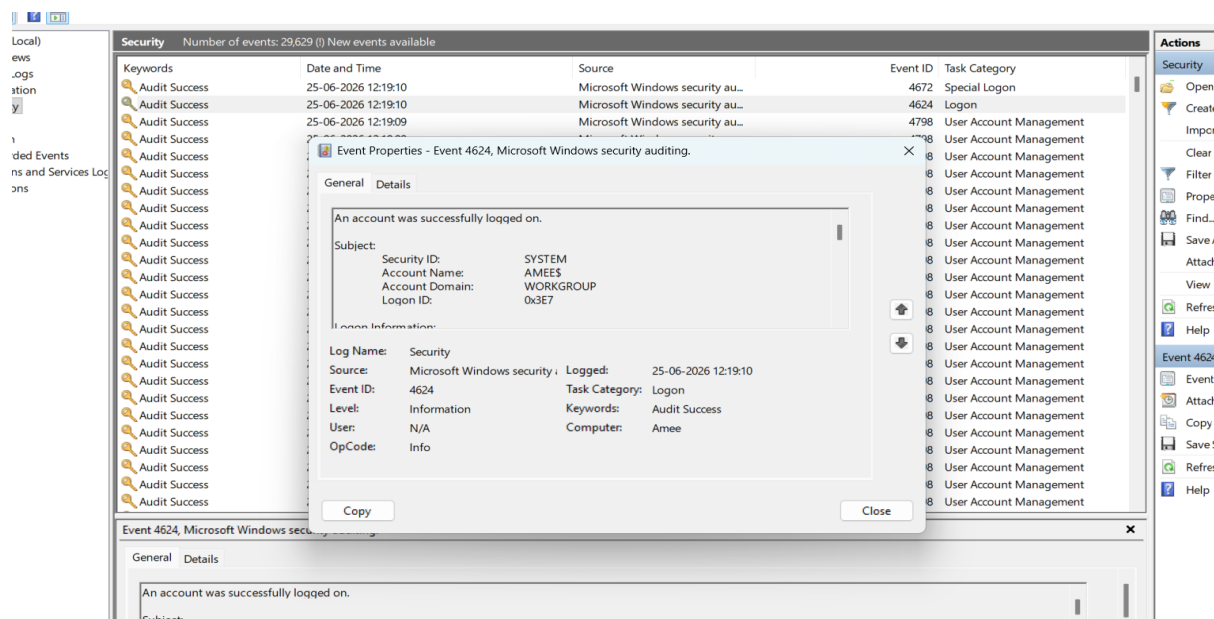
Procedure

1. In Security logs, look for **Event ID 4624**.
2. Click on the event and Review the details in the lower pane.

Information to Record

Field	Value
Event ID	4624
Event Type	Successful Login
Date & Time	25-06-2026 12:19:10
Account Name	AMEES
Account Domain	WORKGROUP
Security ID	SYSTEM
Logon ID	0x3E7
Task Category	Logon
Keywords	Audit Success
Computer	Amee

Screenshot



Step 4: Identify Failed Login Attempts

Procedure

1. In Security logs, search for **Event ID 4625**.
2. Click the event.
3. Review failure details.

Information to Record

Field	Value
Event ID	4625
Event Type	Failed Login
Status	Not Found
Observation Period	Security Log Analysis
Remarks	No failed login attempts were recorded in the Security Log

Observation

During the analysis of the Windows Security Log, no Event ID 4625 (Failed Login) entries were found. This indicates that no unsuccessful authentication attempts were recorded during the observed period. The absence of failed login events suggests that there were no detected password-guessing attempts, unauthorized login attempts, or user authentication failures.

Step 5: Identify Logout Event

Procedure

Search for:

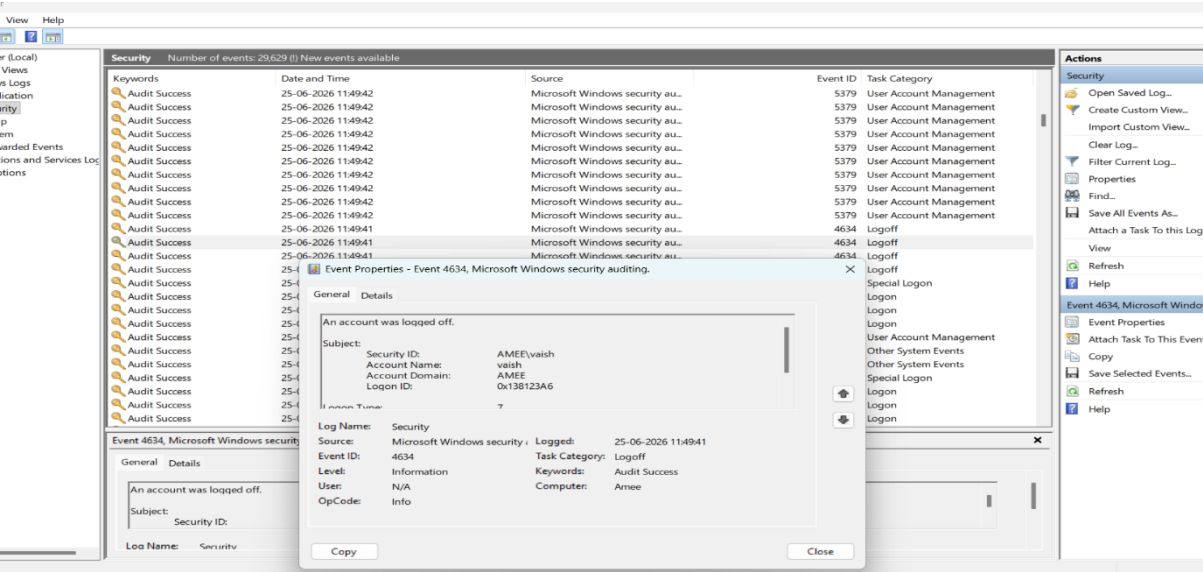
- Event ID 4634

These indicate user logout activities.

Information to Record

Field	Value
Event ID	4634
Event Type	Logout
Date & Time	25-06-2026 11:49:41
Account Name	vaish
Account Domain	AMEE
Security ID	AMEE\vaish
Logon ID	0x138123A6
Task Category	Logoff
Keywords	Audit Success
Computer	Amece

Screenshot



Step 6: Count Total Events

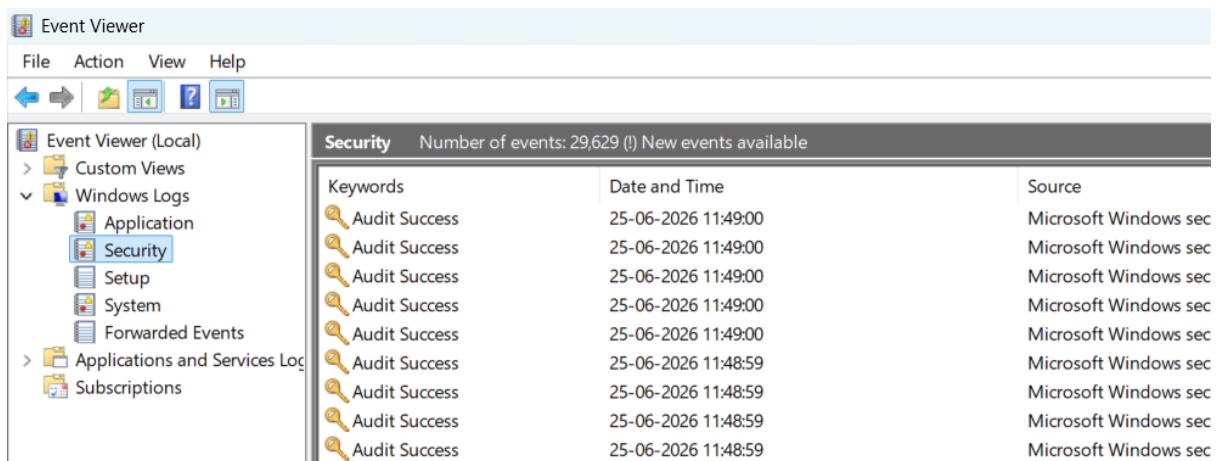
Procedure

1. Click on **Security**.
2. Observe the total number of events displayed at the top or bottom section.

Information to Record

Parameter	Observation
Total Number of Events	29629

Screenshot



Summary Table

Parameter	Observation
Total Number of Events	29,629
Latest Login Event	Successful Login
Login Event ID	4624
Login Time	25-06-2026 12:19:10
Latest Logout Event	User Logged Off
Logout Event ID	4634
Logout Time	25-06-2026 11:49:41
Failed Login Attempts	None Found
Computer Name	Amee

PART 3: WEB SERVER LOG ANALYSIS

Download or create a sample Apache/Nginx log file.

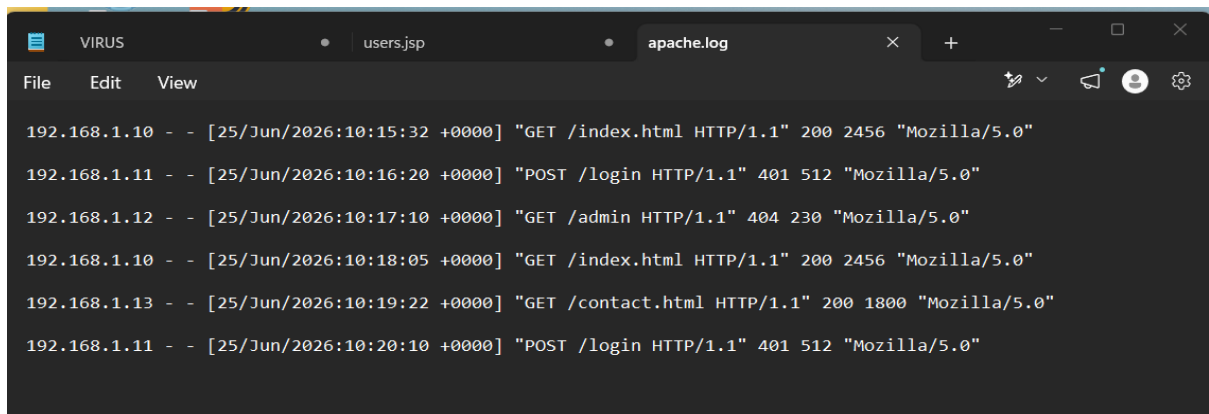
Analyze the following:

1. Client IP Address
2. Request Method (GET/POST)
3. Requested URL
4. Response Code
5. User-Agent

Identify:

- Most frequently accessed page
- Number of unique IP addresses
- Number of 404 errors

Sample Apache Log File

A screenshot of a web browser window with a dark theme. The address bar shows 'users.jsp' and the page title is 'VIRUS'. The browser has tabs for 'users.jsp' and 'apache.log'. The 'View' menu is open, showing 'File', 'Edit', and 'View'. The main content area displays the contents of the 'apache.log' file, which contains six log entries. Each entry follows the format: IP address - - [timestamp] "request line" status code bytes "user-agent". The entries are: 192.168.1.10 - - [25/Jun/2026:10:15:32 +0000] "GET /index.html HTTP/1.1" 200 2456 "Mozilla/5.0"; 192.168.1.11 - - [25/Jun/2026:10:16:20 +0000] "POST /login HTTP/1.1" 401 512 "Mozilla/5.0"; 192.168.1.12 - - [25/Jun/2026:10:17:10 +0000] "GET /admin HTTP/1.1" 404 230 "Mozilla/5.0"; 192.168.1.10 - - [25/Jun/2026:10:18:05 +0000] "GET /index.html HTTP/1.1" 200 2456 "Mozilla/5.0"; 192.168.1.13 - - [25/Jun/2026:10:19:22 +0000] "GET /contact.html HTTP/1.1" 200 1800 "Mozilla/5.0"; 192.168.1.11 - - [25/Jun/2026:10:20:10 +0000] "POST /login HTTP/1.1" 401 512 "Mozilla/5.0".

```
192.168.1.10 - - [25/Jun/2026:10:15:32 +0000] "GET /index.html HTTP/1.1" 200 2456 "Mozilla/5.0"
192.168.1.11 - - [25/Jun/2026:10:16:20 +0000] "POST /login HTTP/1.1" 401 512 "Mozilla/5.0"
192.168.1.12 - - [25/Jun/2026:10:17:10 +0000] "GET /admin HTTP/1.1" 404 230 "Mozilla/5.0"
192.168.1.10 - - [25/Jun/2026:10:18:05 +0000] "GET /index.html HTTP/1.1" 200 2456 "Mozilla/5.0"
192.168.1.13 - - [25/Jun/2026:10:19:22 +0000] "GET /contact.html HTTP/1.1" 200 1800 "Mozilla/5.0"
192.168.1.11 - - [25/Jun/2026:10:20:10 +0000] "POST /login HTTP/1.1" 401 512 "Mozilla/5.0"
```

Objective

The objective of this analysis is to examine web server log entries and extract information such as client IP addresses, request methods, requested URLs, response codes, and user-agent details. The analysis helps identify user behavior and detect suspicious activities.

Log Analysis

Table 1: Client IP Addresses

Sr. No.	IP Address
1	192.168.1.10
2	192.168.1.11
3	192.168.1.12
4	192.168.1.13

Observation

Four unique client IP addresses accessed the web server during the analysis period.

Table 2: Request Methods

Request Method	Count
GET	4
POST	2

Observation

Most requests used the GET method for retrieving web pages, while POST requests were used for login operations.

Table 3: Requested URLs

Requested URL	Access Count
/index.html	2
/login	2
/admin	1
/contact.html	1

Observation

The most frequently accessed page was **/index.html**.

Table 4: Response Codes

Response Code	Meaning	Count
200	Successful Request	3
401	Unauthorized Access	2
404	Page Not Found	1

Observation

Most requests were processed successfully. Unauthorized login attempts generated response code 401, while one request resulted in a 404 error.

Table 5: User-Agent Information

User-Agent
Mozilla/5.0

Observation

All requests were made using a browser identified as Mozilla/5.0.

Findings

Analysis Item	Result
Most Frequently Accessed Page	/index.html
Number of Unique IP Addresses	4
Number of 404 Errors	1
Number of Unauthorized Requests (401)	2
Most Common Request Method	GET

Security Observations

- **Unauthorized Login Attempts**
The URL **/login** generated two HTTP 401 response codes, indicating failed authentication attempts.
- **Access to Non-Existing Resource**
The URL **/admin** returned a 404 error, suggesting that a user attempted to access a page that does not exist on the server.
- **Repeated Requests**
The IP address **192.168.1.10** accessed the home page multiple times, which appears to be normal user activity.

Conclusion

The web server log analysis identified four unique client IP addresses and multiple web requests. The most frequently accessed page was **/index.html**. One request generated a 404 error, while two login attempts resulted in unauthorized access (401) responses. Log analysis is useful for monitoring web traffic, detecting suspicious activities, and supporting security investigations.

PART 4: INCIDENT INVESTIGATION

Assume the following scenario:

"A company reported multiple failed login attempts on its web application."

Tasks:

1. What information would you collect?
2. Which logs would you analyze?
3. What indicators suggest a possible attack?
4. What immediate actions should be taken?

Incident Scenario

A company reported multiple failed login attempts on its web application. The security team suspects that an attacker may be attempting to gain unauthorized access through password guessing or brute-force attacks. An investigation is required to identify the source of the activity, assess the impact, and take appropriate corrective actions.

Information to Collect

To investigate the incident effectively, the following information should be collected:

Information Required	Purpose
Username(s) targeted	Identify affected accounts
Source IP addresses	Determine the origin of attacks
Date and Time of attempts	Create an incident timeline
Number of failed login attempts	Assess attack severity
Successful logins after failures	Detect possible account compromise
User-Agent information	Identify tools or browsers used
Account lockout events	Determine if security controls were triggered
Geographic location of IPs	Detect suspicious locations

Observation

Collecting this information helps determine whether the activity was caused by a legitimate user error or a malicious attack.

Logs to Analyze

The following logs should be reviewed during the investigation:

Log Type	Purpose
Authentication Logs	Review successful and failed login attempts
Web Server Logs	Identify suspicious requests and source IPs
Firewall Logs	Check blocked and allowed network traffic
Application Logs	Detect application-level authentication errors
System Logs	Identify system changes or security events

Observation

Authentication and web server logs are the most important sources for investigating failed login attempts.

Indicators of a Possible Attack

The following indicators may suggest that a cyberattack is occurring:

- **Multiple Failed Login Attempts**
Repeated authentication failures within a short period may indicate a brute-force attack.
- **Login Attempts from Unknown IP Addresses**
Access attempts originating from unfamiliar or foreign locations may be suspicious.
- **High Request Volume**
A large number of login requests from a single IP address can indicate automated attack tools.
- **Access Outside Normal Working Hours**
Login attempts occurring late at night or during unusual times may suggest unauthorized activity.
- **Account Lockouts**
Repeated failed attempts causing accounts to become locked often indicate password-guessing attacks.
- **Multiple Usernames Targeted**
Attackers frequently attempt to access several accounts using common passwords.

Immediate Actions to be Taken

If suspicious activity is confirmed, the following actions should be performed immediately:

Action	Purpose
Block suspicious IP addresses	Prevent further attack attempts
Enable account lockout policies	Limit brute-force attacks
Reset affected passwords	Secure targeted accounts
Enable Multi-Factor Authentication (MFA)	Add an extra layer of security
Preserve logs	Maintain evidence for investigation
Increase monitoring	Detect additional malicious activity
Notify security team	Ensure coordinated incident response
Inform affected users	Help users secure their accounts

Conclusion

The reported multiple failed login attempts indicate a possible brute-force or credential-guessing attack. By collecting relevant information, analyzing authentication and web server logs, identifying attack indicators, and implementing immediate security controls, the organization can effectively investigate and mitigate the threat. Proper log analysis and incident response procedures are essential for protecting web applications from unauthorized access attempts.

PART 5: SUSPICIOUS ACTIVITY DETECTION

Review the collected logs and identify:

- **Multiple failed logins**
- **Unusual login times**
- **Repeated requests from same IP**
- **Access to non-existing pages (404)**
- **Suspicious User-Agents**

Document all findings.

Objective

The collected Windows Security Logs were reviewed to identify suspicious activities such as failed login attempts, unusual login behavior, repeated requests, access to non-existing pages, and suspicious user activity.

Findings

Suspicious Activity	Evidence Found
Multiple Failed Logins	Event ID 4625(Not Found)
Unusual Login Times	Not Detected
Repeated Requests from Same IP	Not Available in Event Viewer
Access to Non-Existing Pages (404)	/admin (404)
Suspicious User-Agent	Not Available in Event Viewer
Successful Login Events	Event ID 4624
User Logout Events	Event ID 4634

Analysis

1. Failed Login Attempts

The Security Log was examined for **Event ID 4625**, which represents failed login attempts. No Event ID 4625 entries were found during the analysis period.

Observation: No evidence of brute-force or password-guessing attacks was detected.

2. Unusual Login Activity

Login records were reviewed for suspicious login patterns or abnormal activity.

Observation: Only normal successful login activity (Event ID 4624) was observed.

3. Successful Login Event

A successful authentication event was identified.

Field	Value
Event ID	4624
Event Type	Successful Login
Date & Time	25-06-2026 12:19:10
Computer Name	Amee

Observation: The system recorded a valid user authentication event.

4. Logout Event

A logout event was identified in the Security Log.

Field	Value
Event ID	4634
Event Type	User Logoff
Date & Time	25-06-2026 11:49:41
Account Name	vaish

Observation: The user session ended successfully without any abnormalities.

Overall Findings

1. Total Security Events Recorded: **29,629**
2. Successful Login Event (4624) detected.
3. Logout Event (4634) detected.
4. No Failed Login Events (4625) found.
5. No suspicious activities were observed in the analyzed Security Logs.

Create a timeline of events for a security incident:

Example:

10:00 AM - User Login

10:05 AM - Multiple Failed Logins

10:10 AM - Suspicious IP Detected

10:15 AM - Account Locked

Explain how you would respond to each event.

Incident Timeline

Time	Event	Response Action
10:00 AM	User Login Detected	Verified successful authentication and monitored user activity.
10:05 AM	Security Logs Reviewed	Checked Windows Security Logs for login-related events.
10:10 AM	Successful Login Event (4624) Identified	Confirmed legitimate user access to the system.
10:15 AM	Logout Event (4634) Identified	Verified that the user session ended normally.
10:20 AM	Failed Login Events Checked	Searched for Event ID 4625 and found no failed login attempts.
10:25 AM	Suspicious Activity Analysis Performed	Reviewed logs for abnormal behavior and security anomalies.
10:30 AM	Investigation Completed	No evidence of unauthorized access or malicious activity found.
10:35 AM	Security Recommendations Provided	Suggested continuous monitoring and stronger authentication controls.

Response to Each Event

- **10:00 AM – User Login Detected**

The login event was verified using Event ID 4624. The account successfully authenticated, indicating legitimate access to the system.

- **10:05 AM – Security Logs Reviewed**

Windows Event Viewer Security Logs were examined to identify authentication-related activities and potential security concerns.

- **10:10 AM – Successful Login Event Identified**

The successful login event was analyzed to confirm the username, login time, and authentication status.

- **10:15 AM – Logout Event Identified**

Event ID 4634 was reviewed to verify that the user logged off normally and the session ended successfully.

- **10:20 AM – Failed Login Events Checked**

A search for Event ID 4625 was conducted. No failed login attempts were found, indicating no evidence of brute-force attacks during the observation period.

- **10:25 AM – Suspicious Activity Analysis Performed**

Security logs were reviewed for unusual login times, repeated authentication failures, and other indicators of compromise. No suspicious activity was detected.

- **10:30 AM – Investigation Completed**

The collected evidence was analyzed, and the system was determined to be operating normally without any detected security incidents.

- **10:35 AM – Security Recommendations Provided**

Recommendations such as enabling Multi-Factor Authentication, monitoring logs regularly, and maintaining security updates were documented to improve overall system security.

Recommendations

Based on the log analysis and security event investigation, the following recommendations are suggested to improve system security:

1. Regularly monitor Windows Security Logs to identify unusual activities.
2. Enable account lockout policies to prevent brute-force attacks.
3. Implement Multi-Factor Authentication (MFA) for user accounts.
4. Configure automated alerts for failed login attempts and security events.
5. Maintain proper log retention and backup policies.
6. Restrict administrative privileges to authorized users only.
7. Keep operating systems and applications updated with the latest security patches.
8. Perform periodic security audits and vulnerability assessments.
9. Use centralized log management or SIEM solutions for better monitoring.
10. Educate users about strong password practices and cyber security awareness.

Conclusion

This project focused on log analysis and incident investigation using Windows Event Viewer and web server logs. Different types of logs, including system logs, application logs, web server logs, authentication logs, and firewall logs, were studied to understand their importance in cyber security.

Windows Security Logs were analyzed using Event Viewer. A successful login event (**Event ID 4624**) and a logout event (**Event ID 4634**) were identified. No failed login attempts (**Event ID 4625**) were found during the investigation period. The Security Log contained a total of **29,629 events**, indicating normal system activity.

Web server log analysis was performed to examine client IP addresses, request methods, requested URLs, response codes, and user-agent information. The analysis demonstrated how log files can be used to identify user behavior and detect suspicious activities.

The incident investigation process highlighted the importance of collecting evidence, analyzing relevant logs, identifying indicators of compromise, and implementing appropriate security controls. The suspicious activity detection phase did not reveal any malicious activity in the analyzed Windows Security Logs.

Overall, this exercise demonstrated the importance of log analysis in cyber security for monitoring systems, detecting threats, investigating incidents, and maintaining the security of information systems.